



# HUST

**ĐẠI HỌC BÁCH KHOA HÀ NỘI**  
HANOI UNIVERSITY OF SCIENCE AND TECHNOLOGY

ONE LOVE. ONE FUTURE.





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# FUNDAMENTALS OF OPTIMIZATION

## MODELLING

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# CONTENT

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- Modelling
- Constraint linearization
- Examples

# MODELLING

- Modelling consists of specifying
  - Decision variables
  - Constraints
  - Objective functions
- A problem can be modelled in different ways (how to define variables)
- Take into account the modelling languages of software tools
  - Constraint Programming solvers: constraints can be stated in flexible ways

# MODELLING - EXAMPLE

- A company need to make a production planning for  $n$  products  $1, 2, \dots, n$  using  $m$  materials  $1, 2, \dots, m$  in which
  - To make one unit of product  $i$ , it require  $c(i, j)$  amount of material  $j$ ,  $\forall i = 1, 2, \dots, n$  and  $j = 1, 2, \dots, m$
  - Amount of material  $j$  is  $A(j)$ ,  $\forall j = 1, 2, \dots, m$
  - Revenue when selling a unit of product  $i$  is  $R(i)$ ,  $\forall i = 1, 2, \dots, n$
- Find a production plan such that the total revenue is maximal

# MODELLING - EXAMPLE

- Decision variables
  - $X(i)$ : amount of product  $i$  to be produced in the planning,  $\forall i = 1, 2, \dots, n$
- Constraints
  - $\sum_{i=1}^n c(i, j)X(i) \leq A(j), \forall j = 1, 2, \dots, m$
- Objective
  - $f(X) = \sum_{i=1}^n R(i)X(i) \rightarrow \max$

# CONSTRAINT LINEARIZATION

- Motivation
  - Linear Programming solvers are very efficient
- Examples
  - How to model  $X = \min\{x_1, x_2\}$  ?



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Solution: define an auxiliary binary variable  $y$ , use big constant  $M$

- $x_1 \geq X$
- $x_2 \geq X$
- $X \geq x_1 - M(1-y)$
- $X \geq x_2 - My$

# CONSTRAINT LINEARIZATION

- Examples
  - How to model  $(x = 1) \Rightarrow (z \geq y)$  where  $x$  is binary variable,  $y$  and  $z$  are real variables ?

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  - How to model  $(x = 1) \Rightarrow (z \geq y)$  where  $x$  is binary variable,  $y$  and  $z$  are real variables ?

Solution: use big constant  $M$

- $M(x-1) + y \leq z$

# CONSTRAINT LINEARIZATION

- Example
  - $(x > 0) \Rightarrow (z \geq y)$  in which  $x, y$  and  $z$  are real variables (and  $x \geq 0$ ) ?

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Solution:

- Let  $M$  be a very big constant,
- Introduce a binary variable  $t \in \{0,1\}$ :
  - $t = 1$  indicates that  $x > 0$ , and  $t = 0$  indicates that  $x = 0$
- Equivalent linear constraints
  - $x \leq M.t$
  - $z + (1-t)M \geq y$

- **Balanced Course Assignment Problem**

- At the beginning of the semester, the head of a computer science department  $D$  have to assign courses to teachers in a balanced way. The department  $D$  has  $m$  teachers  $T=\{0, 2, ..., m-1\}$  and  $n$  courses  $C=\{0, 2,..., n-1\}$ .
  - Each teacher  $t \in T$  has a preference list which is a list of courses he/she can teach depending on his/her specialization. The preference information is represented by a 0-1 matrix  $A_{m \times n}$  in which  $A(t,c) = 1$  indicates that teacher  $t$  can teach the course  $c$  and  $A(t,c) = 0$ , otherwise
  - We know a set  $B$  of pairs of conflicting two courses that cannot be assigned to the same teacher as these courses have been already scheduled in the same slot of the timetable.
  - The load of a teacher is the number of courses assigned to her/him. How to assign  $n$  courses to  $m$  teacher such that each course assigned to a teacher is in his/her preference list, no two conflicting courses are assigned to the same teacher, and the maximal load among teachers is minimal.

# EXAMPLES

- Balanced Course Assignment Problem

| Course  | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
|---------|---|---|---|---|---|---|---|---|---|---|----|----|----|
| credits | 3 | 3 | 4 | 3 | 4 | 3 | 3 | 3 | 4 | 3 | 3  | 4  | 4  |

| Teachers | Preference Courses    |
|----------|-----------------------|
| 0        | 0, 2, 3, 4, 8, 10     |
| 1        | 0, 1, 3, 5, 6, 7, 8   |
| 2        | 1, 2, 3, 7, 9, 11, 12 |

Conflict courses

|   |    |
|---|----|
| 0 | 2  |
| 0 | 4  |
| 0 | 8  |
| 1 | 4  |
| 1 | 10 |
| 3 | 7  |
| 3 | 9  |
| 5 | 11 |
| 5 | 12 |
| 6 | 8  |
| 6 | 12 |

# EXAMPLES

- Balanced Course Assignment Problem

| Course  | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 |
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| credits | 3 | 3 | 4 | 3 | 4 | 3 | 3 | 3 | 4 | 3 | 3  | 4  | 4  |

| Teachers | Preference Courses    |
|----------|-----------------------|
| 0        | 0, 2, 3, 4, 8, 10     |
| 1        | 0, 1, 3, 5, 6, 7, 8   |
| 2        | 1, 2, 3, 7, 9, 11, 12 |

| Teacher | Assigned courses | Load |
|---------|------------------|------|
| 0       | 2, 4, 8, 10      | 15   |
| 1       | 0, 1, 3, 5, 6    | 15   |
| 2       | 7, 9, 11, 12     | 14   |

Conflict courses

|   |    |
|---|----|
| 0 | 2  |
| 0 | 4  |
| 0 | 8  |
| 1 | 4  |
| 1 | 10 |
| 3 | 7  |
| 3 | 9  |
| 5 | 11 |
| 5 | 12 |
| 6 | 8  |
| 6 | 12 |



- **Balanced Course Assignment Problem: A Constraint Programming model**
  - Decision variables
    - $X(i)$ : teacher assigned to course  $i$ ,  $\forall i \in C$ , domain  $D(X(i)) = \{t \in T \mid A(t,i) = 1\}$
    - $Y(i)$ : load of teacher  $i$ , domain  $D(Y(i)) = \{0,1,\dots,n-1\}$
    - $Z$ : maximum load among teachers
  - Constraints
    - $X(i) \neq X(j), \forall (i,j) \in B$
    - $Y(i) = \sum_{j \in C} (X(j) = i), \forall i \in T$
    - $Z \geq Y(i), \forall i \in T$
  - Objective function to be minimized:  $Z$

- **Balanced Course Assignment Problem: An Integer Linear Programming model**
  - Decision variables
    - $X(i,j) = 1$ : teacher  $i$  is assigned to course  $j$ , and  $X(i,j) = 0$ , otherwise,  $\forall i \in T, j \in C$ , domain  $D(X(i,j)) = \{0,1\}$
    - $Y(i)$ : load of teacher  $i$ , domain  $D(Y(i)) = \{0,1,...,n\}$
    - $Z$ : maximum load among teachers
  - Constraints
    - $\sum_{i \in T} X(i,j) = 1, \forall j \in C$
    - $X(t,i) + X(t,j) \leq 1, \forall (i,j) \in B, t \in T$
    - $Y(i) = \sum_{j \in C} X(i,j), \forall i \in T$
    - $Z \geq Y(i), \forall i \in T$
  - Objective function to be minimized:  $Z$

- **Travelling Salesman Problem (TSP)**

- A salesman departs from point 1, visiting  $N-1$  points 2, ...,  $N$  and comes back to the point 1. The travelling distance from point  $i$  and point  $j$  is  $d(i,j)$ ,  $i,j = 1,..., N$ . Compute the route of minimal total travelling distance

# EXAMPLES

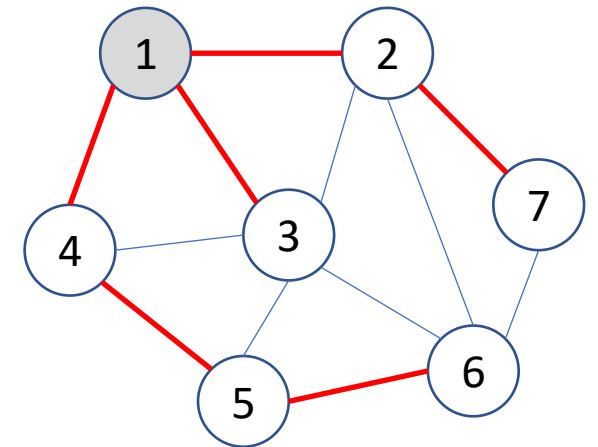
- Travelling Salesman Problem (TSP): An Integer Linear Programming model
  - Decision variables
    - Binary variable  $X(i,j) = 1$  if the route traverses from point  $i$  to point  $j$ , and  $X(i,j) = 0$ , otherwise.
  - Constraints
    - $\sum_{j=1}^N X(i,j) = \sum_{j=1}^N X(j,i) = 1, \forall i \in \{1,2,\dots,N\}$
    - $\sum_{(i,j) \in S} X(i,j) \leq |S| - 1, \forall S \subseteq \{1,2,\dots, N\} \text{ and } |S| < N$
  - Objective function to be minimized

$$f(X) = \sum_{j=1}^N \sum_{i=1}^N d(i,j)X(i,j)$$

# EXAMPLES

- **MultiCast Routing Problem**

- Given a network  $V = \{1, \dots, N\}$  is the set of nodes,  $E \subseteq V^2$  is the set of links between nodes. A node  $s \in V$  is the source node which will transmit a package to others nodes. A node receiving the package can continue transmit this package to adjacent nodes.
  - $t(i,j)$  and  $c(i,j)$  are transmission time and transmission cost when transmitting the package from node  $i$  to node  $j$
- Compute the set of links used for broadcasting the package from the source node to all other nodes such that
  - Total transmission time from  $s$  to any node cannot exceed a given value  $L$
  - Total transmission cost is minimal

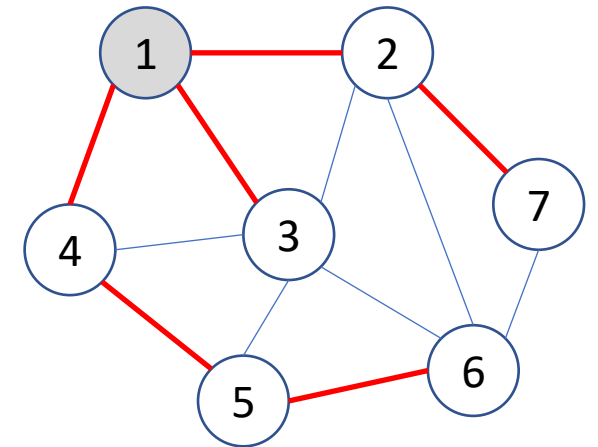


# EXAMPLES

- **MultiCast Routing Problem**

- Denote  $A(i) = \{j \in V \mid (i,j) \in E\}$ ,  $M$  is a big constant
- Decision variables
  - Binary variable  $X(i,j) = 1$  if the package is transmitted from node  $i$  to node  $j$ , and  $X(i,j) = 0$ , otherwise,  $\forall (i,j) \in E$
  - $Y(i)$ : time-point when the package arrives at node  $i$ ,  $\forall i \in V$
- Constraints
  - $\sum_{i \in A(j)} X(i,j) = 1, \forall j \in V \setminus \{s\}$
  - $Y(i) + t(i,j) + M(1-X(i,j)) \geq Y(j), \forall (i,j) \in E$
  - $Y(i) + t(i,j) + M(X(i,j)-1) \leq Y(j), \forall (i,j) \in E$
  - $Y(i) \leq L, \forall j \in V \setminus \{s\}$
  - $Y(s) = 0$
- Objective function to be minimized

$$f(X) = \sum_{(i,j) \in E} c(i,j)X(i,j)$$



- **Facility Location Problem**

- There are  $M$  sites  $1, 2, \dots, M$  that can be used to open facility for servicing  $N$  customers  $1, 2, \dots, N$ .
  - $f(i)$  is the cost for opening the site  $i$
  - $Q(i)$  is the capacity of site  $i$  (maximum amount of good it can serve customers)
  - $c(i,j)$  is the cost for transporting unit of good from site  $i$  to customer  $j$
  - $d(j)$  is the total demand (amount of goods) of customer  $j$
- Compute a planning (which site to be opened and amount of good each opened site serves a customer) such that
  - Capacity constraint is satisfied
  - Total cost is minimal

- **Facility Location Problem**

- Decision variables

- $Y(i)$  – binary variable,  $Y(i) = 1$  means that the site  $i$  is opened, and  $Y(i) = 0$ , otherwise
    - $X(i,j)$  – amount of good site  $i$  serves customer  $j$

- Constraints

- $\sum_{i=1}^M X(i,j) = d(j), \forall j = 1, \dots, N$
    - $\sum_{j=1}^N X(i,j) \leq Q(i)Y(i), \forall i = 1, \dots, M$
    - $0 \leq X(i,j) \leq d(j)Y(i), \forall i = 1, \dots, M, \forall j = 1, \dots, N$

- Objective functions

$$\sum_{i=1}^M f(i)Y(i) + \sum_{i=1}^M \sum_{j=1}^N c(i,j)X(i,j) \rightarrow \min$$



A graphic on the left side of the slide featuring a dark blue background with a large, stylized circular pattern of red dots of varying sizes, creating a sense of depth and movement. The word "HUST" is centered within this pattern.

**HUST**

**THANK YOU !**